

SC2006 Software Engineering

Lecture 20 Supplement

Conrad Watt

Lt19 recap

- Black box vs white box testing
- Control flow graphs
- Coverage concepts

Plan for the rest of the semester

- Thursday 10th April
 - revision lecture
- Monday 14th April
 - example past paper questions
- Thursday 17th April
 - non-assessed case study lecture (TBC)

Linearly independent paths?

- Concept of basis path testing focusses on identifying tests which exercise all the “linearly independent paths”
- What does this mean?

The idea

- A collection of paths is linearly independent if each path passes through at least one node that the others do not
- A basis path-based test suite seeks to create a test suite that's ***as large as possible*** and covers every node, where each test exercises a linearly independent path (with respect to the other tests)

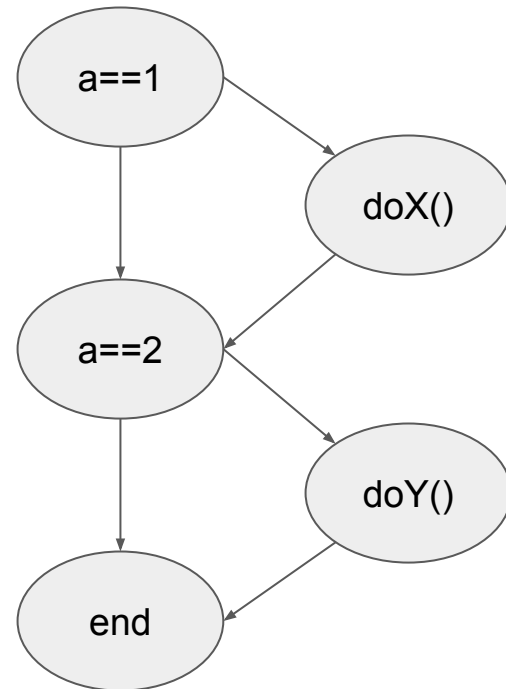
The idea

- A collection of paths is linearly independent if each path passes through at least one node that the others do not
- A basis path-based test suite seeks to create a test suite that's ***as large as possible*** and covers every node, where each test exercises a linearly independent path (with respect to the other tests)
- The formulae in the slides give an upper bound to the number of possible tests / linearly independent paths
 - this upper bound is known as the ***cyclomatic complexity***

Example

e.g.

```
if (a == 1) {  
    doX();  
}  
if (a == 2) {  
    doY();  
}
```

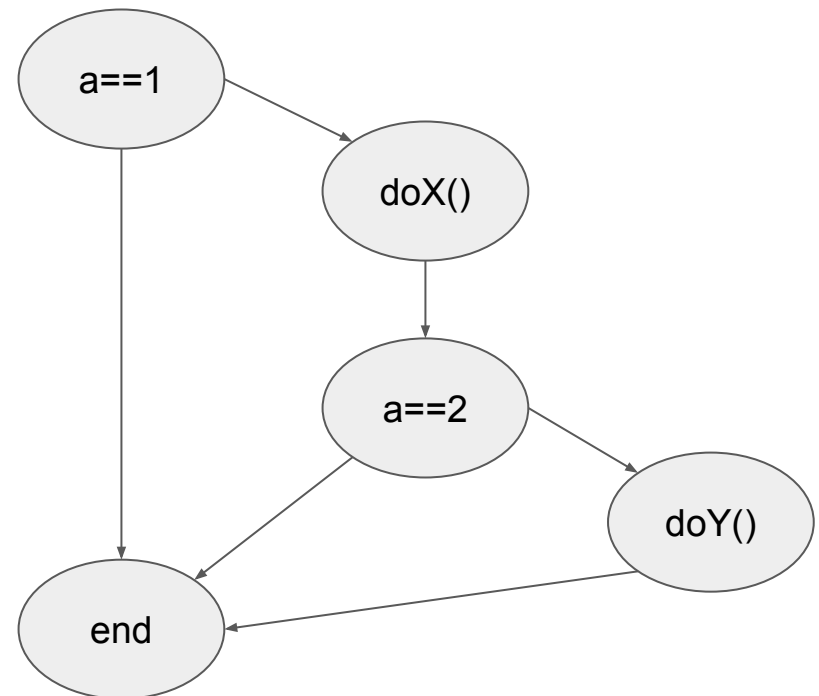


The gap between cyclomatic complexity and test suite size

- This is caused by unreachable code

e.g.

```
if (a == 1) {  
    doX();  
    if (a == 2) {  
        doY();  
    }  
}
```



The gap between cyclomatic complexity and test suite size

- cyclomatic complexity can be seen as a measure of the trustworthiness of a function
- even if many paths are unreachable (and therefore can't be tested), guaranteeing this takes a lot of effort

e.g.

```
if (a == 1) {  
    doX();  
    if (very complicated condition equivalent to a == 2) {  
        doY();  
    }  
}
```